Soldering of Electrical Terminals – General Data

1. General

This standard specifies the procedures and requirements for soldering wires to electrical terminals.

2. Materials and Equipment

A. Materials

- Solder as specified in Table 1 — Solder and Flux Selection
- Liquid flux as specified in Table 1 — Solder and Flux Selection
- Isopropyl Alcohol (IPA) to Federal Specification TT–I–735, Grade A or B

Table 1 Solder and Flux Selection

WIRE TO BE SOLDERED	SOLDER (Note)	FLUX
tin plated wire or silver plated wire	Sn60 or Sn60Pb40 (60% Tin/40% Lead)	Rosin base liquid flux to MIL-F-14256 Type R or J-STD-004 Type ROM1
	Sn63 or Sn63Pb37 (63% Tin/37% Lead)	
	"No clean" flux core solder Sn63Pb37 flux type LR to QQ-S-571 (e.g., Interflux USA Inc. IF 14 solder wire or Kester Solder Co. Kester #245 solder wire)	"No clean" liquid flux to MIL-F-14256 Type LRL (e.g., Interflux USA Inc. IF-2005M or Kester Solder Co. 951 flux)
nickel plated wire	Sn96 or Sn96.3Ag3.7 (96.3% Tin/3.7% Silver)	Rosin base liquid flux to MIL-F-14256 Type RA
Note The designation of solder has been revised in QQ-S-571 to more clearly describe the components making up the solder. For example, the previous designation Sn60 has been changed to Sn60Pb40 for solder that is 60% Tin and 40% Lead. It is acceptable to use solder described using either designation.		

B. Equipment

- Soldering equipment capable of heating the solder joint to the required temperature without damaging the wire, terminal, or wire insulation. Refer to Table 2 — Selection of Soldering Irons for recommended soldering iron sizes.
- Soldering wick
- Solder extractor
- Cotton gloves

Table 2 Selection of Soldering Irons

WIRE GAUGE (AWG)	SOLDERING IRON
26 - 20	35 watt
18 - 14	50 watt
12 - 10	60 watt, hatchet
8 - 6	150 - 300 watt
Note If a relatively small gauge wire is being soldered onto a larger terminal a more powerful soldering iron may be needed.	

3. Safety Precautions

   A. Take care to ensure that personnel are not accidentally burned by the hot iron or solder. Place hot irons on the bench in the holders provided when not in use.

   B. Do not eat or drink in soldering areas. Always wash your hands after soldering, especially before eating.

   C. Take care to avoid inhaling the fumes generated when soldering.

4. Requirements

   A. Maintain strict cleanliness when soldering.

   B. Apply a small amount of solder. A correctly soldered joint with minimum solder is stronger and more efficient then a solder joint that has too much solder.

   C. Adjust the heat application so that a minimum amount of time is required to solder the joint. If insufficient heat is applied, it will take longer for the solder to flow properly which can cause the wire lead insulation to burn.

   D. Do not force cool the solder joint. Force cooling can cause pits or fractures in the solder joint.

   E. Flux helps the flow of the solder and prevents dewetting or non-wetting of the solder joint through its cleaning action. The flux is activated at a temperature just below the temperature required for the solder to flow so it is important to wait until the solder flows freely before soldering the joint. Not allowing the solder to flow freely before soldering the joint can result in cold solder joints, dewetting, non-wetting or rosin or flux joints.

   F. Do not disturb the solder joint until it is cooled. Disturbing the solder joint before cooling can cause a weaker and inefficient joint.

   G. Always handle wire harnesses and wire assemblies with care to avoid damage such as fractured soldered joints.

5. Soldering of Wires

   A. Wire Stripping and Pre-Solder Cleaning

      1) Strip the appropriate length of wire insulation to expose the conductor.

      2) Before hook up of wires, remove any scale or dirt on terminals using a fine abrasive. After abrasive cleaning, solvent clean terminals with a bristle brush and IPA solvent.

      3) After solvent cleaning, dry the terminal by wiping with a tissue. When solvent washing in the area of DIP switches, relays, etc., take care to avoid contact of the solvent with the components.

      4) Before tinning, solvent clean wire leads with a bristle brush and IPA to remove any oil or grease (e.g. finger oils). After solvent cleaning, dry the wire lead by wiping with a tissue.

      5) Do not touch cleaned terminals or wires with bare fingers. Wear clean cotton gloves to avoid contamination.

   B. Tinning

      1) Tin the soldering iron tip by heating the iron to maximum temperature and then contacting the tip with rosin cored solder before wiping the iron with a single motion on a water dampened sponge.

2) Tin all stripped wire leads before soldering.

a) Tin all stripped wire leads using a soldering iron as follows:

1 Place the wire in a clamp and brush liquid flux along the stripped length to within approximately 1/16" of the wire insulation. Alternatively, it is acceptable to dip the wire end in a pot of liquid flux, but only to within approximately 1/16" of the wire insulation.

2 Place the solder and the soldering iron approximately halfway along the length of the stripped length, and wipe towards the insulation to within approximately 1/8" before reversing direction and wiping off the end of the wires.

b) Dip tin wire ends as follows:

1 Brush liquid flux onto the end of the stripped end of the wire to within approximately 1/16" of the insulation. Alternatively, it is acceptable to dip the wire end in a pot of liquid flux, but only to within approximately 1/16" of the wire insulation.

2 Dip the fluxed end of the wire into the solder pot to within approximately 1/8" of the insulation.

C. General Requirements for Mechanical Hook Up of Terminals

1) Allow wires to curve smoothly to their place of attachment with no sharp bends.

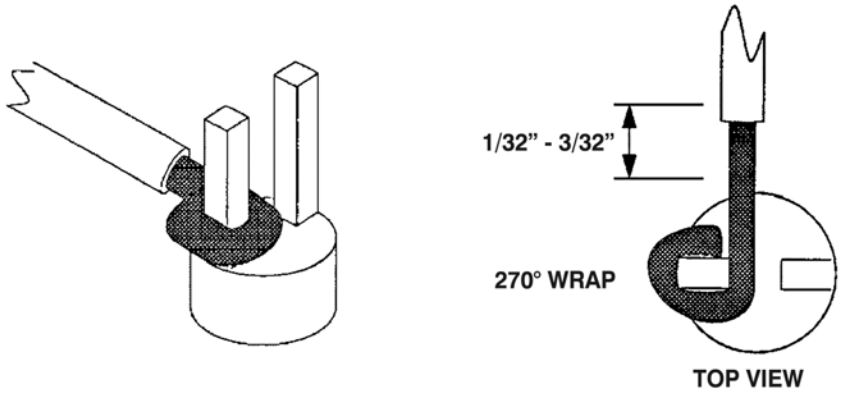
2) Position the larger leads below the smaller leads if soldering various sizes of wire leads to the same terminal.

3) Use smooth jawed, round nosed pliers to form the wire around the terminal to the proper turn position. Ensure to locate the end of the insulation 1/32" - 3/32" away from the terminal.

4) Ensure that the connection is mechanically secure to prevent movement during the soldering operation.

D. Mechanical Hook Up of Bifurcated Terminal Assemblies

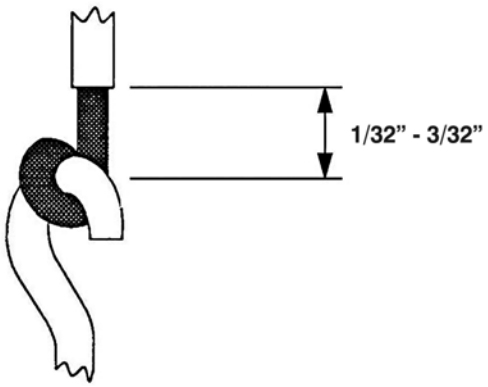
1) Wrap the wire lead approximately 270° around the fork (refer to Figure 1). If soldering two or more wire leads to one bifurcated terminal, wrap the wires alternately on each fork.



Mechanical Hook-Up of Bifurcated Terminals
Figure 1

E. Mechanical Hook Up of Hook terminal Assemblies

1) Wrap the wire lead approximately 270° around the hook as shown in Figure 2.



Mechanical Hook Up of Hook Terminals
Figure 2

F. Mechanical Hook Up of Pierced or Eyelet Terminal Assemblies

1) Wrap the wire end approximately 270° through the eyelet (refer to Figure 3).

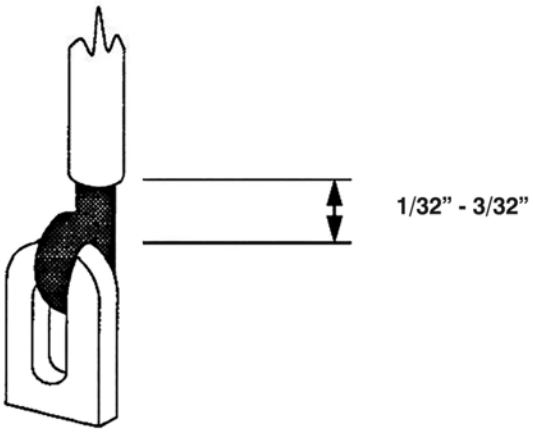
2) If connecting two wires to a single pierced or eyelet terminal and the slot or hole in the terminal is not large enough to admit both wires, hook up the wires as follows:

a) Strip one of the wires back an additional 1/2" (approximately).

b) Wrap the other wire around the stripped back portion.

c) Maintain an insulation gap of 1/32" - 3/32" for both wires.

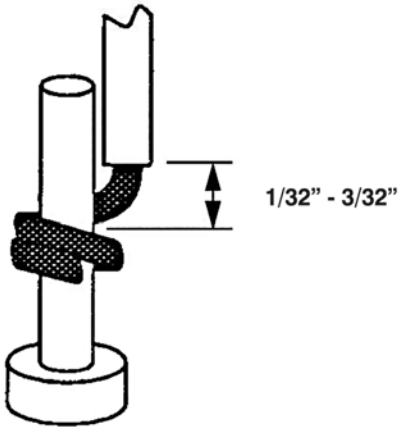
d) Wrap the protruding wire end approximately 270° through the eyelet.



Mechanical Hook Up of Eyelet Terminals
Figure 3

G. Mechanical Hook Up of Pin Terminal Assemblies

1) When soldering small gauge wire to pin terminals, wrap the wire lead once or twice (360° - 720°) around the pin to ensure a mechanically sound connection (refer to Figure 4).



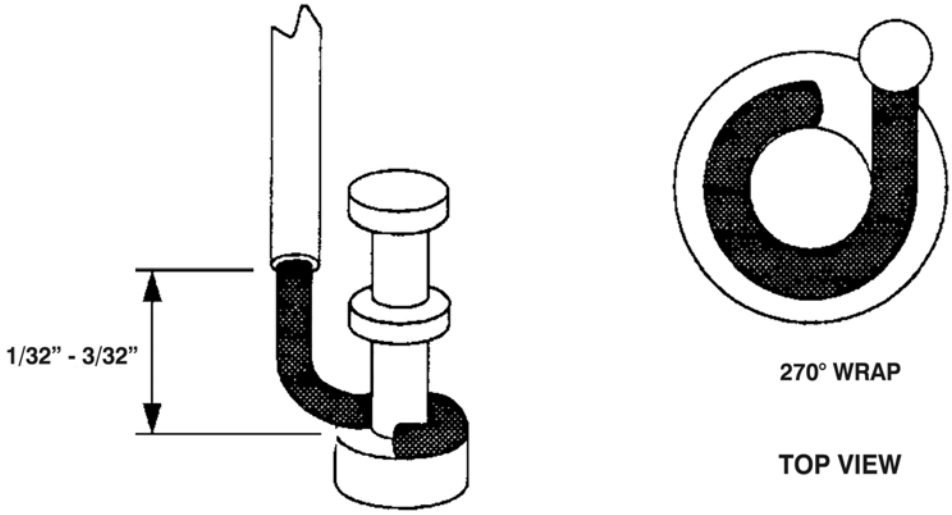
Mechanical Hook Up of Pin Terminals
Figure 4

H. Mechanical Hook Up of Turret Terminal Assemblies

1) Wrap wire ends approximately 270° around the post without extending below the terminal base or flange (refer to Figure 5).

2) Hook up the wire lead at the base/lower post junction or similarly at the top shoulder/upper post junction.

3) All wire leads on a single terminal shall wrap in the same direction with the larger leads connected below the smaller leads.



Mechanical Hook Up of Turret Terminals
Figure 5

I. General Soldering Practices

1) If specified that the joint should be over-sleeved with heat shrinkable tubing, insert the wire into a suitable length of heat shrinkable tubing and slide the tubing along the wire out of the way. After the solder joint has been completed and cooled, shrink the heat shrinkable tubing over the joint.

2) Just before soldering, apply a small amount of liquid flux to the terminal and the tinned wire lead junction.

3) Wipe the soldering iron tip, with a single motion, on a water dampened sponge just before making each connection. This will remove the oxide film which forms on the tip of the soldering iron. Refer to Table 3 — Soldering Iron Tips for Terminal Applications for the recommended type of soldering tip.

Table 3 Soldering Iron Tips for Terminal Applications

TERMINAL TYPE	RECOMMENDED SOLDERING IRON TIP
bifurcated	pyramid or short taper chisel
hook	blunt or conical
pierced eyelet	narrow semi-chisel or pyramid
pin	pyramid or short taper chisel
solder cup	chisel or screw driver
turret	chisel or screw driver

4) Place the soldering iron tip on the terminal near the terminal and lead wire junction so that the majority of the heat is applied to the larger mass of the terminal. Rest the soldering iron tip on the terminal without exerting any pressure. Place the soldering iron tip on the lowest portion of the terminal, if possible. It is often helpful to form a solder bridge right after placing the soldering iron tip on the joint to aid heat transfer. Ensure that there is no movement of the conductor when the tip is applied to the connection. Keep the soldering iron as far away from the wire insulation as possible.

5) The joint has reached the proper temperature to add solder when the solder of the solder bridge melts and flows into the joint.

6) Apply solder as soon as the solder of the solder bridge melts and flows into the joint. Apply solder to the part of the connection furthest away from the soldering iron tip. "Paint" the solder onto the joint lightly, using only enough solder to cover the joint, leaving the wire strands visible and creating a slightly concave fillet with a feather edge.

7) Use a soldering iron and tip that will allow completion of the connection within 5 seconds. Hold the heating time to a minimum to prevent damage to insulation or components and to avoid burning or carbonizing of flux. Use heat sinks to prevent damage to heat sensitive components, if necessary. Do not exceed 5 seconds of heat application, at one time, to any single connection.

8) Take care not to damage adjacent components, previously installed, when soldering. When soldering wires to miniature connectors, take care to prevent damage to insulation.

9) If soldering eyelet or pierced terminals, apply only sufficient solder to cover the wire and form a fillet between the terminal and the wire on both sides. It is not necessary to fill the eyelet with solder.

10) Remove the solder strand from the joint before removing the iron.

11) Do not handle or disturb the connection until the solder has completely cooled. If necessary, use supporting and locating clamps to hold wires in position while soldering and cooling.

12) Do not force cool the solder joint by blowing, fanning or any other means.

13) If soldering connections to closely spaced terminals, it is often easier to use Pulse Heat Resistance Tweezers.

14) If soldering unsecured terminal assemblies, use a holding fixture to avoid handling shock to the joint, if possible.

J. Soldering of Cup Terminal and Co-Axial Cable Pin Assemblies

1) Solder cup terminals and coaxial cable pin assemblies as follows:

a) Place a length of tinned wire lead in the cup, ensuring that the wire end bottoms in the cup, and trim level with the top of the cup. Use the length of this trimmed piece to determine the proper length, including insulation gap, for the wire lead.

b) Trim and insert a length of solder into the cup. Insert enough solder so that the solder and wire lead will fill the cup.

c) Place the resistance tweezers, or soldering iron, on the cup and heat the solder until it melts and then insert the wire end, ensuring the wire end bottoms in the cup.

d) Before removing the heat, agitate the wire end in the cup to release any entrapped flux.

e) Allow the joint to cool.

K. Post Soldering Cleaning

After the solder has cooled, solvent clean the solder joint using a stiff bristle brush and IPA to remove all traces of soldering flux. Take care when solvent cleaning in the area of DIP switches, relays, etc., to avoid contact of the solvent with the components. After solvent cleaning, dry the solder joint with a clean tissue.

6. Removal of Solder

If solder connections require repair, remove the solder from the joint using a solder extractor or a solder wick.

7. Inspection/Check

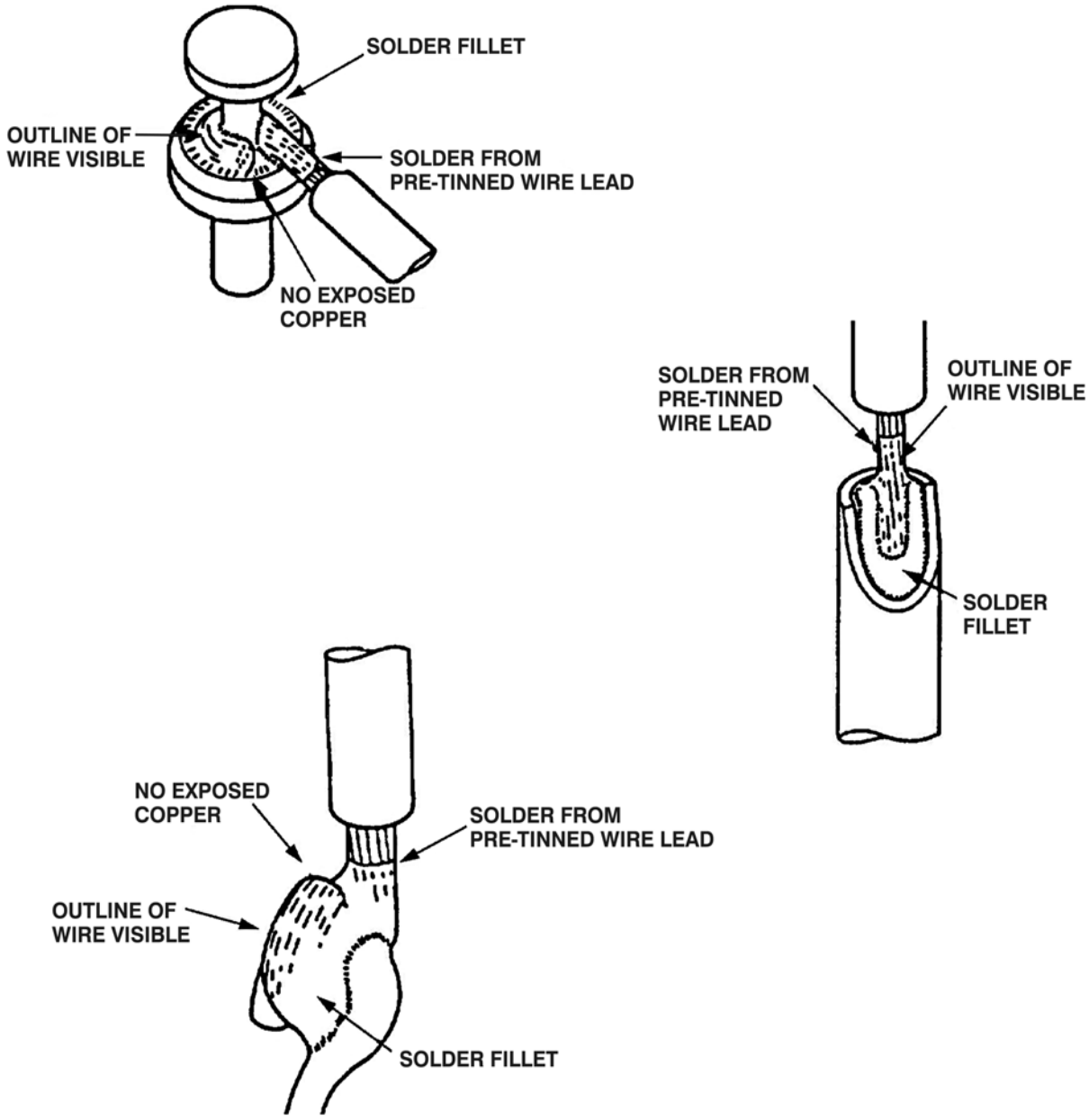
A. Soldering and wiring shall be uniform in appearance.

B. The hook up of wires to terminals shall be as specified in this standard. Insufficient or excessive anchoring shall be cause to reject the joint.

C. Acceptable solder joints (refer to Figure 6) shall have a shiny surface, show a wetting action on all surfaces of the solder joint, a slightly concave fillet and a feather edge. Stranded wires shall be tinned as specified herein before soldering onto terminals or into solder cups and co-axial pins.

D. Clearance between the solder joint and the wire insulation shall be 1/32" - 3/32".

E. Evidence of any of the defects listed in Table 4 — Solder Defects shall be cause to reject the affected solder joint.



Acceptable Solder Joints
Figure 6

Table 4 Solder Defects

DEFECT	DESCRIPTION	USUAL CAUSES	PREVENTATIVE ACTION
cold solder	- the solder usually appears dull and crystalline - the solder may be shiny and smooth - solder appears stacked or piled up	- underheated solder - joint disturbed during cooling - improper solder flow - improper wetting action	- increase heat - increase dwell time - do not handle the joint or components during the cooling cycle
disturbed solder	- the solder has a dull, porous and crystalline appearing surface - under magnification the solder may reveal cracks or fractures	- handling shock during the cooling cycle - movement between the lead and the terminal during the soldering operation (i.e. spring-back)	- allow the solder to fully solidify before handling - ensure that the mechanical connection is tight before soldering - avoid exerting excessive pressure with the soldering iron tip during the soldering operation
rosin or flux joint	- the solder usually appears dull and crystalline - the solder may be shiny and smooth - solder appears stacked or piled up - joint shows evidence of rosin entrapped within the joint - condition results from an underheated solder joint; the flux has not been heated sufficiently to produce the cleaning action - due to poor flow and wetting action raw flux (underheated) is trapped within the joint and will seep out	- insufficient heat applied	- ensure joint elements are heated sufficiently to melt and flow the solder freely (flux will activate just before solder flow)
insufficient solder	- the terminal or lead shows exposed base metal in the area of the solder joint - the absence of a solder fillet between the terminal and the conductor	- improper solder flow - improper wetting action - improper application of heat - improper application of flux - improper application of solder - insufficient solder applied to the joint	- improve method of applying heat, flux or solder

Table 4 Solder Defects (continued)

DEFECT	DESCRIPTION	USUAL CAUSES	PREVENTATIVE ACTION
excessive solder	- contour of conductor is completely obscured - solder overflow beyond the confines of the area being soldered	- insufficient heat, allowing the solder to solidify too rapidly - improper solder flow - improper wetting action - excessive solder applied to the joint	- improve method of applying heat, flux or solder
non-wetting	- non-wetting is distinguished from the surrounding areas of solder coated material by exhibiting its original bare uncoated base metal - condition produced when molten solder contacts the metal surface and recedes with none of the solder adhering to it	- improper flux application - improper solder application - improper heat application	- improve method of applying flux, heat and solder - ensure that solder joint elements and tools are clean before soldering - increase amount of flux applied to joint
dewetting	- the solder appears dull and depleted in contrast to the shiny and more thickly coated surrounding areas. Condition produced when molten solder contacts the metal surface and recedes, leaving only a thin film of solder on it	- improper flux application - improper solder application - improper heat application	- reduce dwell time during heat application - ensure that solder joint elements and tools are clean before soldering - increase amount of flux applied to the joint
wicking	- solder extending under the lead insulation - some wicking is normal but should not distort or damage the insulation - wicking of solder under the insulation of a length more than 1/4" is cause to reject the affected joint	- wicking is the result of capillary action drawing the solder up between the strands of the lead - excessive heat, dwell time or solder	- apply only enough solder to bind the wire strands when tinning stranded connectors and only enough to wet the connection and form a light fillet when soldering terminal connections
fractured solder joint	- the solder has cracked or broken between the joint elements	- mechanical stresses during the cooling cycle - thermal stresses during the cooling cycle - stresses due to design application	- allow the solder to fully solidify before handling - allow the solder to solidify without forced cooling such as blowing or fanning

Table 4 Solder Defects (continued)

DEFECT	DESCRIPTION	USUAL CAUSES	PREVENTATIVE ACTION
overheated solder	- the solder has a chalky, dull or crystalline appearance - solder shows evidence of coarse grain porosity or pitting	- excessive heat - excessive dwell time	- reduce heat - reduce dwell time
pits	- small holes occurring as imperfections in the surface of the solder which penetrate entirely through the solder to the base metal	- contaminants or dross on the soldering iron tip - flux and oxide contaminants on the solder surface - too rapid cooling	- ensure that the soldering iron tip has been properly tinned - ensure that the surfaces to be soldered are clean and all surface oxides have been removed - do not force cool the solder joint
pin holes	- small holes occurring as imperfections in the surface of the solder which penetrate entirely through the solder to the base metal	- contaminants and oxides on the surface to be soldered - failure of flux to clean surface - dirty or improperly tinned soldering iron tip	- ensure that the surfaces to be soldered are clean and all surface oxides have been removed - increase flux application or use a more active flux - ensure that the soldering iron tip has been properly tinned and is kept clean
solder projections	- solder projections (i.e. icicles, spikes, bridges, etc.) are undesirable protrusions from a solidified solder joint	- insufficient heat causing the solder to "follow tip" when removed from the connection - insufficient dwell time resulting in solder elements which are not hot enough to flow the solder properly - excessive solder applied	- ensure that the soldering iron and tip are big enough for the job - ensure that the soldering iron tip is properly tinned and cleaned - ensure that the surfaces to be soldered are hot enough to melt and flow the solder properly before applying the solder - apply only enough solder to wet and bond the connection

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